Ex. No: 05

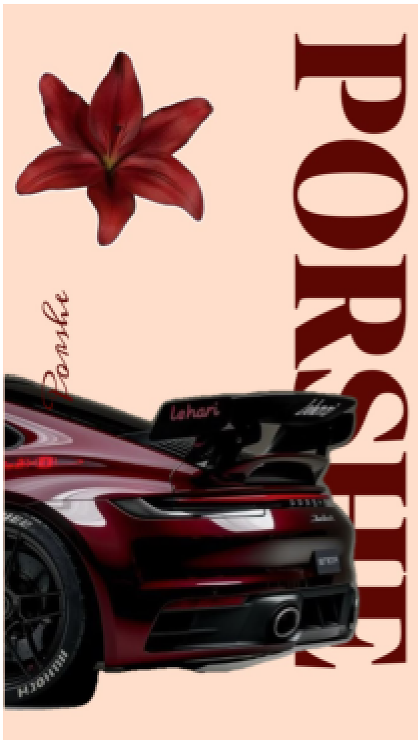
Implementation of filter

Name: ASWINI D **Reg. No:** 212225240015 **Slot:** T1-H15

In [2]:
```
#import the Libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

In [3]:
```
#Read the Input Image
image = cv2.imread("PIC (2).jpg")
```

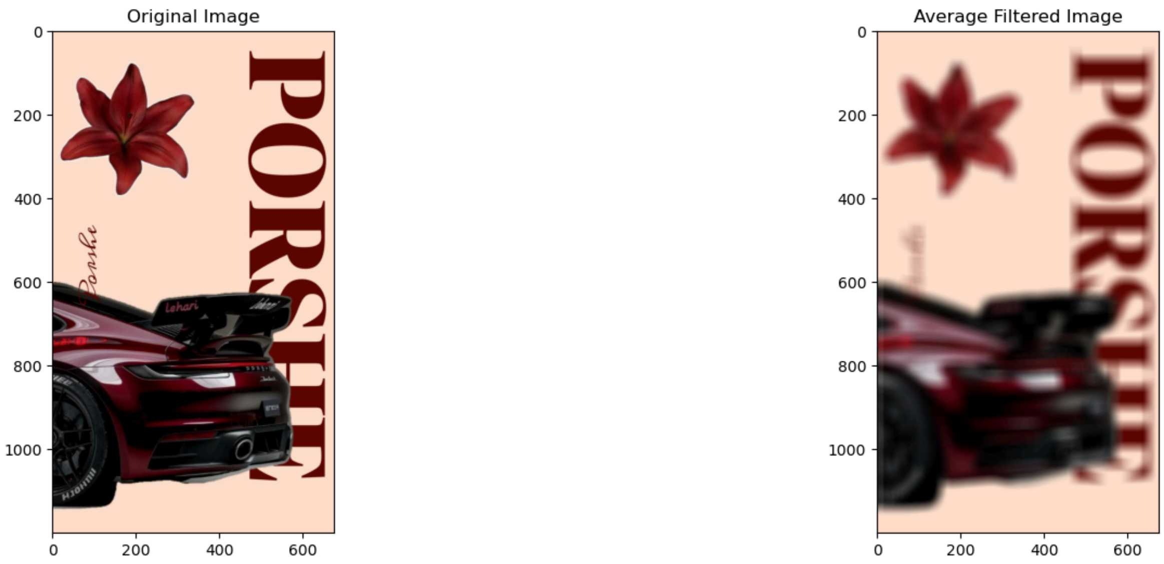
In [4]:
```
#Convert the image from BGR to RGB format for display.
img1 = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
plt.imshow(img1)
plt.axis('off')
plt.show()
```



In [5]:
```
#Average Filter
avg_filter = cv2.blur(img1, (30,30))
```

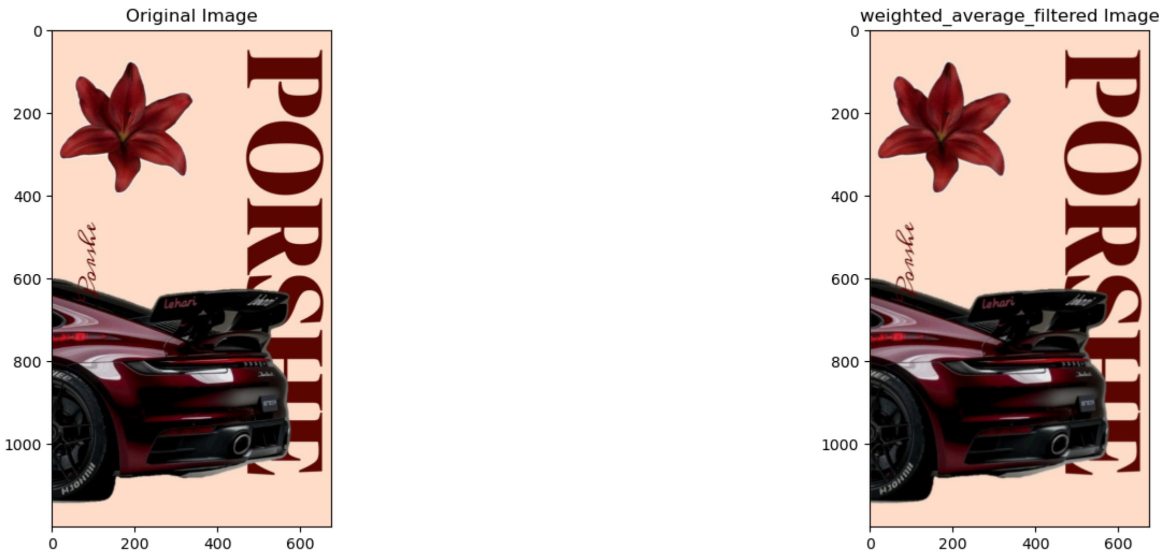
In [6]:
```
plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(avg_filter); plt.title('Average Filtered Image')
```

Out[6]:
```
Text(0.5, 1.0, 'Average Filtered Image')
```



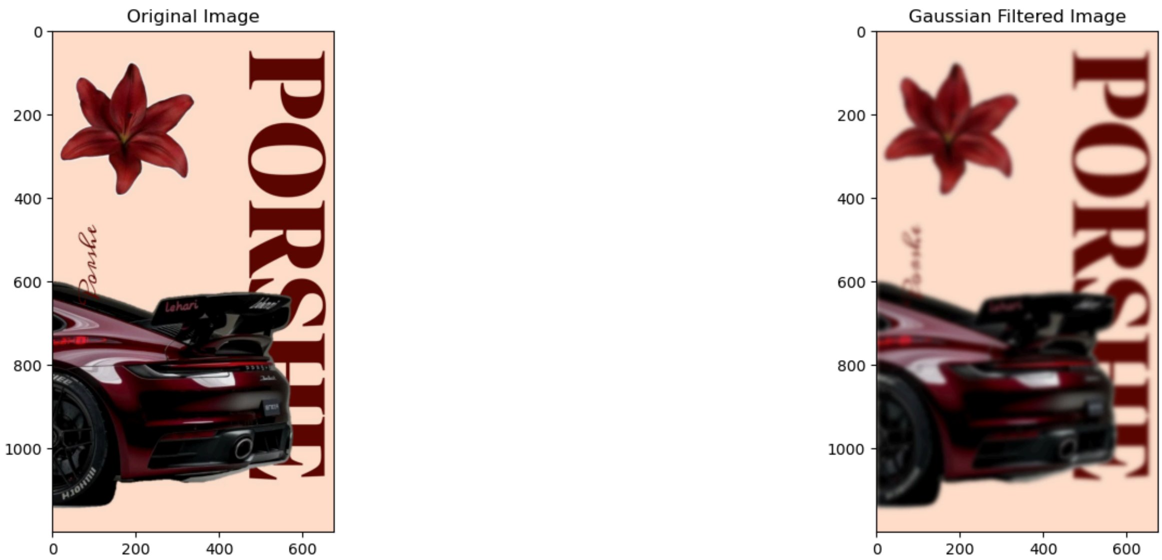
In [7]:
```
#Weighted Average
kernel = np.array([[1,2,1],
                   [2,4,2],
                   [1,2,1]])/16
weighted_avg_filter = cv2.filter2D(img1, -1, kernel)
```

In [8]:
```
plt.figure(figsize = (18, 6))
plt.subplot(121);plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122);plt.imshow(weighted_avg_filter); plt.title('weighted_average_fi
plt.show()
```



In [10]:
```
#Gaussian Filter
gaussian_filter = cv2.GaussianBlur(img1, (29,29), 0, 0)
```

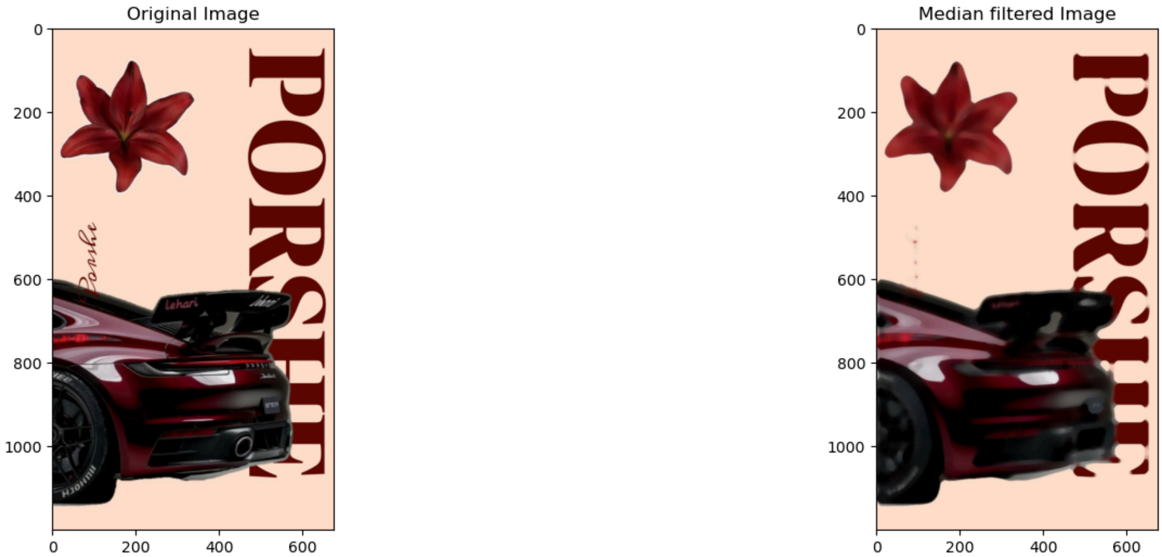
In [11]:
```
plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(gaussian_filter); plt.title('Gaussian Filtered Imag
plt.show()
```



In [12]:
```python
#Median Filter
median_filter = cv2.medianBlur(img1, 19)
```

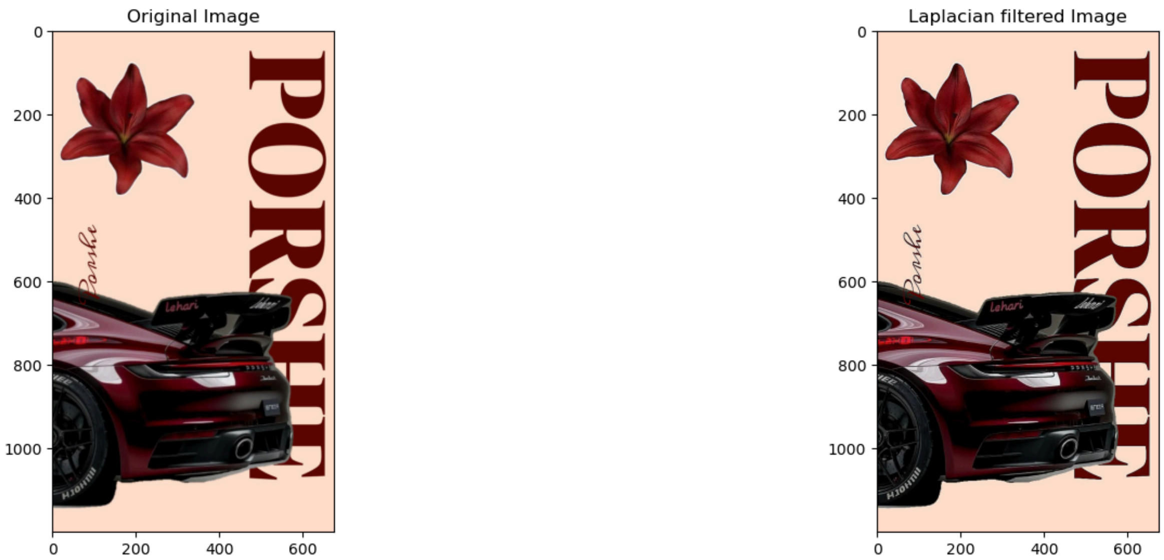
In [13]:
```python
plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(median_filter); plt.title('Median filtered Image')
```

Out[13]:
```
Text(0.5, 1.0, 'Median filtered Image')
```



In [14]:
```python
#Laplacian Sharpening
laplacian_kernel = np.array([[0, -1, 0],
                            [-1, 5, -1],
                            [0, -1, 0]])
sharpened_laplacian_kernel = cv2.filter2D(img1, -1, kernel = laplacian_kernel)
```

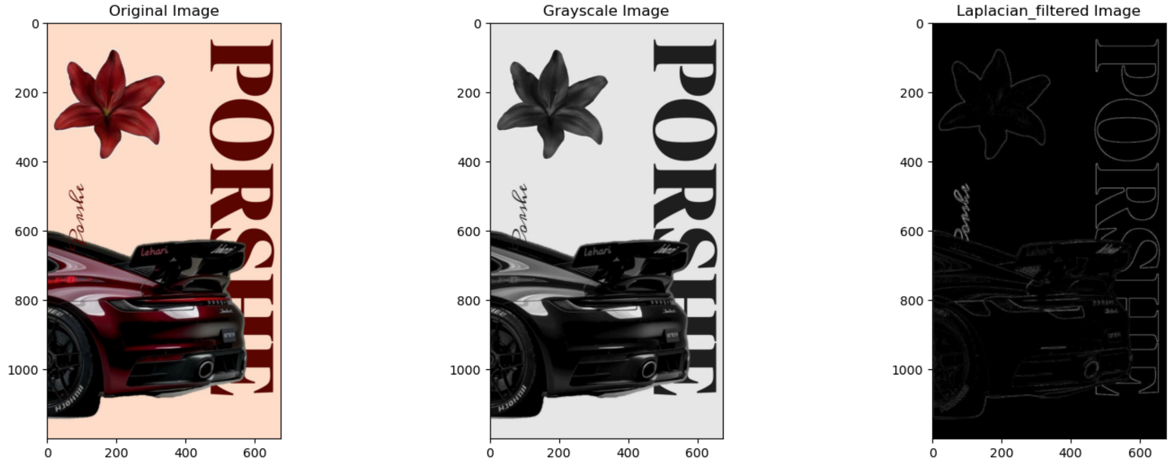
In [15]:
```python
plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(sharpened_laplacian_kernel); plt.title('Laplacian f
plt.show()
```



In [16]:
```python
#Laplacian filter for Grayscale Image
gray_image = cv2.cvtColor(img1, cv2.COLOR_RGB2GRAY)
laplacian_operator = cv2.Laplacian(gray_image, cv2.CV_64F)
laplacian_operator = np.uint8(np.absolute(laplacian_operator))
```

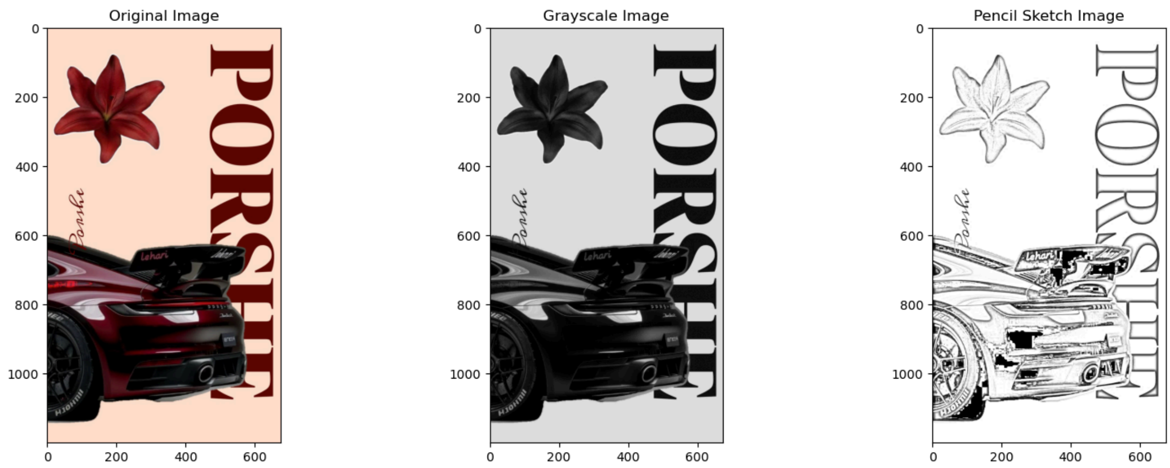
In [17]:
```python
plt.figure(figsize = (18, 6))
plt.subplot(131); plt.imshow(img1); plt.title('Original Image')
plt.subplot(132); plt.imshow(gray_image, cmap='gray'); plt.title('Grayscale Imag
plt.subplot(133); plt.imshow(laplacian_operator,cmap='gray'); plt.title('Laplaci
```

Out[17]: Text(0.5, 1.0, 'Laplacian_filtered Image')



In [18]:
```python
#Pencil Sketch
# Convert to grayscale
gray = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY)
# Invert the grayscale image
invert = cv2.bitwise_not(gray)
# Apply Gaussian blur
blur = cv2.GaussianBlur(invert, (21, 21), 0)
# Invert the blurred image
blur = cv2.bitwise_not(blur)
# Create pencil sketch
sketch = cv2.divide(gray, blur, scale=256.0)
# Display images
plt.figure(figsize = (18, 6))
plt.subplot(131); plt.imshow(img1); plt.title('Original Image')
plt.subplot(132); plt.imshow(gray, cmap='gray'); plt.title('Grayscale Image')
```

```
plt.subplot(133); plt.imshow(sketch,cmap='gray'); plt.title('Pencil Sketch Image
plt.show()
```

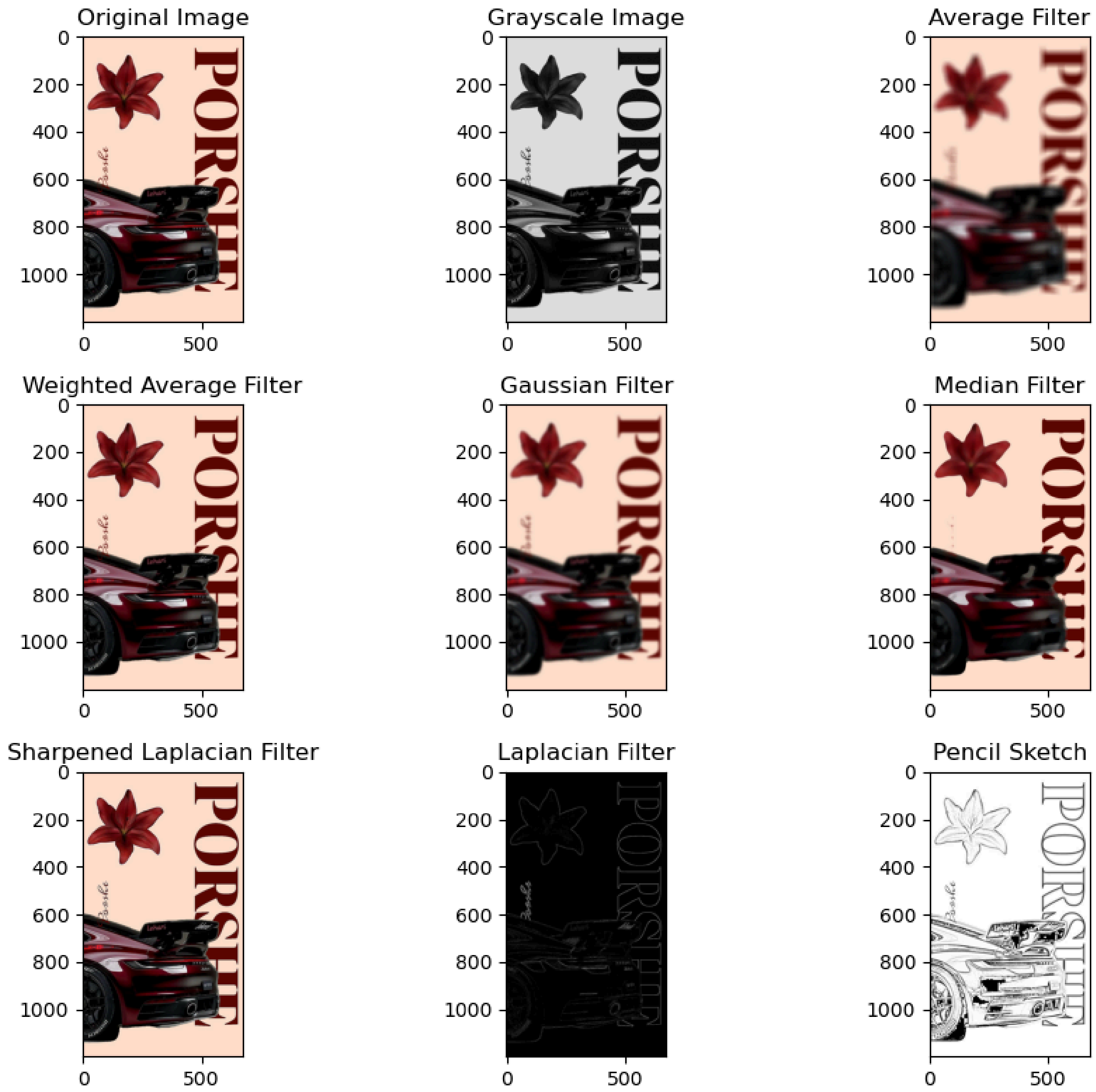


In [19]:

```
#All Images in a grid for a comparision
plt.figure(figsize=(10, 8))

plt.subplot(3,3,1)
plt.imshow(img1);plt.title("Original Image")
plt.subplot(3,3,2)
plt.imshow(gray, cmap="gray");plt.title("Grayscale Image")
plt.subplot(3,3,3)
plt.imshow(avg_filter);plt.title("Average Filter")
plt.subplot(3,3,4)
plt.imshow(weighted_avg_filter);plt.title("Weighted Average Filter")
plt.subplot(3,3,5)
plt.imshow(gaussian_filter);plt.title("Gaussian Filter")
plt.subplot(3,3,6)
plt.imshow(median_filter);plt.title("Median Filter")
plt.subplot(3,3,7)
plt.imshow(sharpened_laplacian_kernel);plt.title("Sharpened Laplacian Filter")
plt.subplot(3,3,8)
plt.imshow(laplacian_operator,cmap="gray");plt.title("Laplacian Filter")
plt.subplot(3,3,9)
plt.imshow(sketch,cmap="gray");plt.title("Pencil Sketch")

plt.tight_layout()
plt.show()
```


Original Image
Grayscale Image
Average Filter
Weighted Average Filter
Gaussian Filter
Median Filter
Sharpened Laplacian Filter
Laplacian Filter
Pencil Sketch

```
In [ ]:
```